
Three questions, and what the tables now answer

Look-ahead depth · optimizer family · the oracle instrument

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Exp3 · Llama-3.2-1B therapist · gpt-4o-mini patient + oracle · Claude Haiku 4.5 held-out judge · 96 personas, persona-paired

The thesis asks exactly three questions

RQ-i

Does looking ahead help?

K = 0 vs K = 5, within each optimizer

Answered — depends on the optimizer

RQ-ii

Which optimizer?

PTO vs GRPO, at matched K and matched compute

Answered — depends on K

RQ-iii

Does the oracle questionnaire change the conclusion?

Q1+Q2 vs WAI-SR vs CSQ-8 vs MI-SAT / MITI

Held — not yet bought

Everything after this slide is one of these three, plus what it cost and what to decide.

One slide of setup, then straight to the questions

- **Therapist.** Llama-3.2-1B, bf16, LoRA $r=16$. Trained by PTO (preference tree + DPO) or GRPO (group-relative), $K \in \{0, 5\}$ look-ahead turns.
- **Patient + training oracle.** gpt-4o-mini. 96 fixed personas; every contrast below is paired on persona identity, $n = 96$.
- **Held-out judge.** Claude Haiku 4.5 — different model family, never played the patient, never was the training reward.
- **Training reward.** Q1 + Q2 only (matching the ICLR paper). Evaluation reads all 8 instruments.

THE GRID, AS OF TODAY

4 arms PTO/GRPO $\times K \in \{0, 5\}$

40 model states 11 + 11 + 11 + 7 (base + iterations)

2 graders every state scored on both

30,720 cells 40 $\times$ 8 rubrics $\times$ 96 personas, per grader

Complete except GRPO $K=5$, which stopped at iteration 6 (slide 16).

RQ-i

Does looking ahead help?

Scoring a reply on its own rewards openings that look good in isolation. Scoring it after K simulated turns rewards openings that lead somewhere.

The lever works — but it points in opposite directions

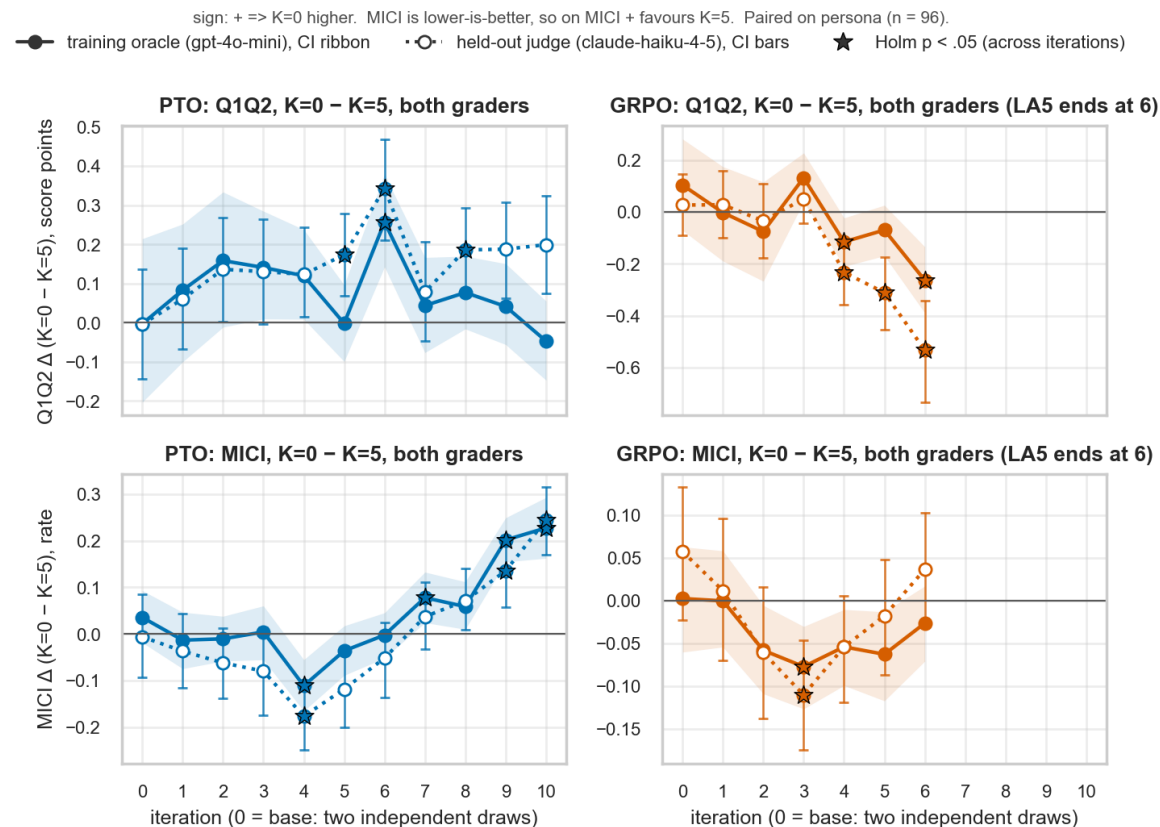
VERDICT
Look-ahead HELPS GRPO and HURTS PTO. At iteration 6 both effects are significant, on both graders.

iteration	PTO · primary	PTO · held-out	GRPO · primary	GRPO · held-out
4	+0.120 (0.20)	+0.123 (0.21)	-0.115 (-0.25) *	-0.233 (-0.37) **
5	-0.002 (-0.00)	+0.173 (0.33) *	-0.070 (-0.13)	-0.311 (-0.43) **
6	+0.257 (0.42) ***	+0.343 (0.51) ***	-0.263 (-0.42) ***	-0.533 (-0.55) ***

Q1+Q2, persona-paired mean difference (Cohen's dz). Sign: + = K=0 higher. * p_holm < .05 ** < .01 *** < .001

NOT THIS
Not that look-ahead is worthless. On PTO it never significantly WINS on either grader — that is a null, not a harm, everywhere except iteration 6.

The same lever, both optimizers, both graders



READ IT AS

Two panels that mirror each other. Whatever look-ahead is doing, it is not a property of the lever alone — it is an interaction with the optimizer.

New this week: the training grader can now see it too

VERDICT

The $K \times$ optimizer interaction is now significant on the PRIMARY oracle, not just the held-out judge.

iter	DiD · primary	dz	p_holm	DiD · held-out	dz	p_holm
4	+0.235	0.286	.104 (ns)	+0.356	0.401	.008
5	+0.068	0.095	1.000 (ns)	+0.484	0.525	<.001
6	+0.520	0.605	<.001	+0.876	0.754	<.001

- **Until iteration 5**, the interaction was visible only to the held-out judge — which was the single sharpest argument in the thesis for having a second grader.
- **At iteration 6**, the primary sees it as well (dz 0.605). The held-out judge still sees it 1.68× more strongly (0.876 / 0.520).

RETRACTED

“The grader that WAS the training reward cannot see this effect.” True through iteration 5; false at 6. The defensible claim is now “sees it less sharply”.

Does the gain survive a grader that was never the reward?

VERDICT
On GRPO, K=5 keeps 84% of its gain under the held-out judge; K=0 keeps 57%. Confidence intervals disjoint.

method	iter	K=0 retention	95% CI	K=5 retention	95% CI	disjoint?
GRPO	6	0.567	[0.379, 0.728]	0.844	[0.745, 0.959]	yes
PTO	10	0.823	[0.720, 0.947]	0.639	[0.551, 0.738]	no

- **Retention** = (gain measured by the held-out judge) ÷ (gain measured by the training oracle). 1.0 means the improvement is fully real to a grader with no stake in it.
- **Reward hacking, quantified.** A low retention number is a gain the training oracle believes and an independent grader does not.

NOT THIS
Not that PTO reward-hacks and GRPO does not. PTO's K=0 arm retains 0.823 — higher than either GRPO arm. The contrast is WITHIN method, across K.

RQ-ii

Which optimizer?

PTO grows a best-of-M reranked trunk; GRPO slices an on-policy rollout. Same candidate budget ($M = G = 8$), same temperature, same oracle.

At K = 0, PTO wins clearly

VERDICT
At the matched 10-iteration endpoint with no look-ahead, PTO beats GRPO on both graders, with a large effect on the held-out judge.

metric	Δ primary	dz	p_holm	Δ held-out	dz	p_holm
Q1+Q2	+0.507	0.729	<.001	+0.609	1.265	<.001
MITI (MI integrity)	+0.352	0.459	<.001	+0.253	0.648	<.001
MICI (inconsistency) ↓	-0.346	-0.989	<.001	-0.225	-0.667	<.001

- **Sign.** Δ = PTO – GRPO. Positive favours PTO; MICI is lower-is-better, so a negative Δ also favours PTO.
- **Mechanism.** GRPO K=0 peaks at iteration 8 (4.082) then regresses to 3.753 while its MI-inconsistency climbs. PTO climbs steadily to 4.260.

At K = 5, the answer flips

VERDICT

Turn look-ahead on and GRPO wins. At matched iteration 6 it beats PTO on both graders; best-vs-best it ties on the primary and WINS on the held-out judge.

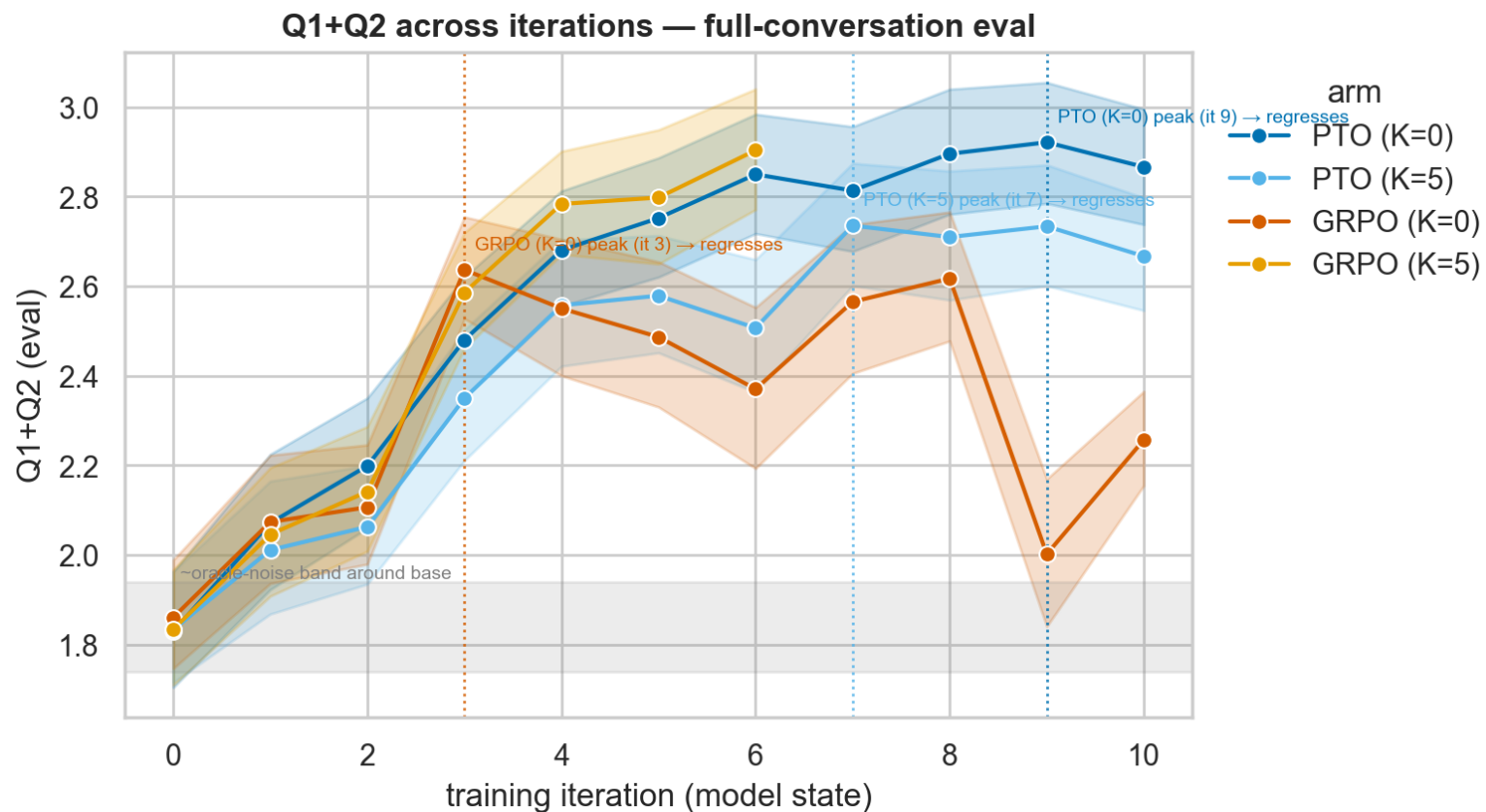
comparison	Δ primary	dz	p_holm	Δ held-out	dz	p_holm
matched iteration 6	-0.332	-0.437	<.001	-0.397	-0.599	<.001
best vs best	+0.078	0.133	1.000 (ns)	-0.168	-0.352	.009

- Δ = PTO – GRPO, so every negative number on this slide favours GRPO.
- **Best-vs-best pairs** PTO@10 against GRPO@6 (primary) and PTO@7 against GRPO@6 (held-out) — each arm at its own peak.
- **So RQ-i and RQ-ii are not separable.** “Which optimizer is better” has no answer that is not conditional on K.

NOT THIS

Not that GRPO overtakes PTO given enough iterations. GRPO K=5 has SIX; PTO has ten. The flip is at matched iteration, and the arm is censored.

The whole picture, held-out judge



READ IT AS

GRPO K=0 (orange) peaks early then collapses to 2.257. GRPO K=5 climbs monotonically to 2.903 in six iterations — the highest endpoint of any arm on this grader.

An iteration is not a unit of spend

VERDICT

On the compute axis PTO dominates outright: it reaches iteration 10 for 8.1 GPU-h against GRPO's 27.9 — 3.4× cheaper, and it scores higher.

arm	iterations	GPU-h	GPU-h / iter	Q1+Q2 primary	Q1+Q2 held-out
PTO K=0	10	8.12	0.81	4.260	2.866
PTO K=5	10	19.68	1.97	4.307	2.667
GRPO K=0	10	27.91	2.79	3.753	2.257
GRPO K=5	6 (+ partial 7)	30.53	5.09	4.229	2.903

- **Why PTO is cheap.** Its dominant phase is building preference pairs (5.7 of 8.1 h), which does not scale with the optimizer step count. GRPO computes its reward inside the training loop.
- **The pair that is no longer matched.** The two GRPO arms were “budget-matched within 3%” at iteration 5. At iteration 6 the K=5 arm is 9.4% MORE expensive; that framing is retired.

RQ-iii

Does the oracle questionnaire change the conclusion?

STATUS — HELD, DELIBERATELY

All four arms train on Q1+Q2 only. Every state is EVALUATED on all 8 instruments, so we can already say the conclusions do not depend on which rubric we READ.

What is untested is whether they depend on which rubric we TRAIN ON. That needs new training runs, not new scoring — four more arms at roughly 8–28 GPU-h each.

Recommendation: keep it held. RQ-i and RQ-ii are the thesis; this is a fourth chapter we do not need.

The measurement thread, in one slide

VERDICT

An independent grader, in a different model family, reproduces 88.5% of all 6,240 arm × metric contrasts — and 99.5% of the large ones.

subset	n	same sign
all contrasts	6,240	88.5%
$ \Delta \geq 0.10$	4,035	94.5%
$ \Delta \geq 0.25$	2,539	97.4%
$ \Delta \geq 0.50$	1,309	99.5%

— **6,240** = 8 metrics × C(40, 2) = 8 × 780.

- **Oracle repeatability.** ICC(2,1) 0.86–0.99 across four re-draws of the same conversations.
- **Dependability (1 judge).** Strong for Q1/Q2/WAI-SR/CSQ-8/PCT (0.91–0.96) and MICI (0.85). MITI is the weak instrument at 0.62 — name it whenever a channel result rests on MITI.
- **Corrected today.** This deck’s predecessors quoted MITI 0.55 / MICI 0.63. Neither figure appears in any table; the rendered values are 0.622 and 0.845.

One honest slide about the run that stopped

WHAT HAPPENED

Iteration 7 needs 106 optimizer steps. 40 are on disk. Across four Colab sessions, 132 further steps were computed and then thrown away.

session	date	steps trained	steps saved	outcome
1	19 Aug	1 → 103	1 → 30	writes to Drive stopped mid-save
2	20 Aug	—	—	OpenAI org spend limit (384 of 395 log lines)
3	20 Aug	—	—	OpenAI org spend limit
4	20 Aug	31 → 99	31 → 40	writes to Drive stopped mid-save

- **Training was never the problem.** Steps after the stall ran at 162.1 s against 161.1 s before it. The GPU and the oracle were both healthy the whole time
- only new files stopped reaching Drive.
- **It is not chronic.** All 16 previously completed iterations have exactly as many saved artifacts as optimizer steps. Iteration 7 is the only anomaly.
- **The fix is operational.** Write checkpoints to local Colab disk and copy to Drive once per iteration. Nothing in the experiment name, the reward or the data changes. Credits are already topped up.

Where the evidence is thin — said out loud

- **Every contested endpoint is a single 96-conversation draw.** The only noise floor we have is at the base: 4 independent draws of the identical base policy give 54 same-policy contrasts, 0 of which reach even uncorrected $p < .05$ (max $|dz|$ 0.128). That is reassuring for the base and says nothing about a trained checkpoint.
- **GRPO K=5 is censored at 6.** Every statement about it is “within six iterations”. Whether its lead persists, or whether it regresses the way GRPO K=0 did after iteration 8, is simply unobserved.
- **MITI dependability is 0.62 off one judge,** and MITI carries the MI-integrity channel results. There is no channel-level ICC at all.
- **All 96 personas are used for both training and evaluation** at every iteration, so every number is in-sample with respect to the patient distribution.
- **Two statistics quoted in earlier decks have no owning table** and are withdrawn until they are rendered: MI acts per 1,000 therapist characters, and the per-conversation over-praise share.

Three things I would like decided today

1

Finish GRPO K=5?

Resume iteration 7 and run to 10: ~16–18 GPU-h, ~\$85–130. The arm lands near 48–50 GPU-h against its K=0 sibling's 28.

My recommendation: yes — it is the only arm without an endpoint, and it is currently the best-scoring arm on the held-out judge.

2

Which axis does the thesis report?

Matched iteration and matched compute answer different questions and disagree at K=5. PTO wins both at K=0.

My recommendation: report both, and name the axis in every sentence that quotes a number.

3

Buy the replicate draw?

A second independent 96-conversation draw of ~5 contested endpoints: ~\$10 and about one A100-hour, or four free hours locally. No code change needed.

My recommendation: yes — it either retires the endpoint-fragility objection thesis-wide or tells us the headline is fragile.

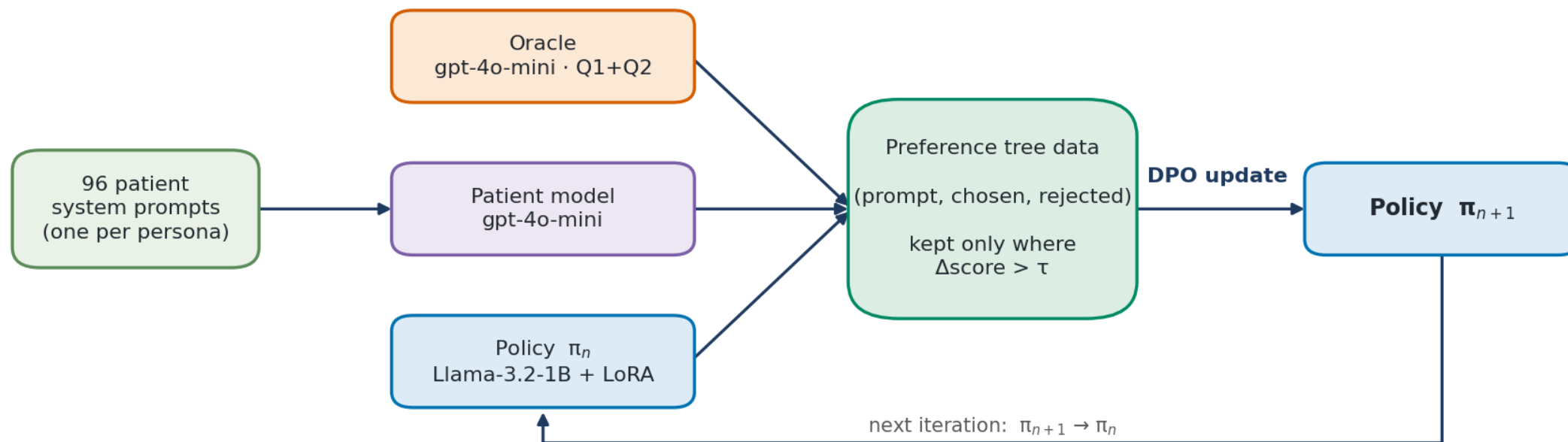
IN ONE LINE

Look-ahead is not a knob with a sign.

It helps group-relative optimization and hurts preference-tree optimization, and which optimizer you should prefer depends on whether you turned it on.

PTO remains 3.4× cheaper and wins outright at $K = 0$. GRPO with look-ahead is the best arm the held-out judge has seen — with six iterations to PTO's ten.

PTO — preference tree + DPO

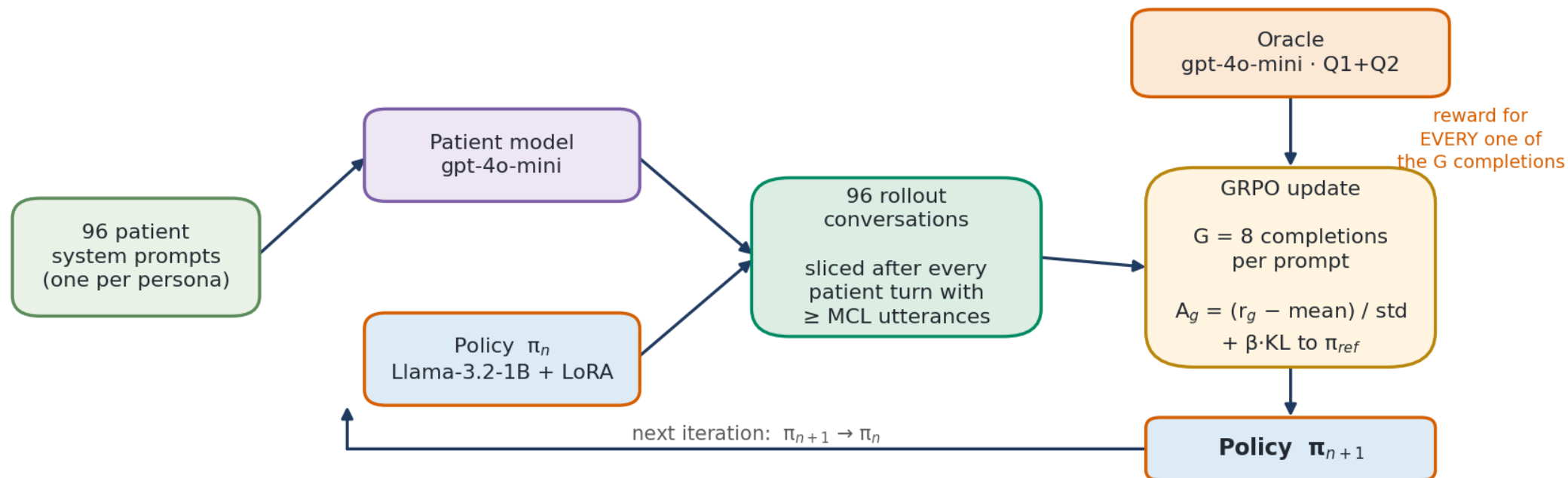


Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.

IN ONE LINE

Branch M candidates per therapist turn, look ahead K, score, keep (chosen, rejected) where the margin clears τ , then a DPO update.

GRPO — group-relative rollout



Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.

As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

IN ONE LINE

Slice the rollout after every patient turn, sample G completions per prompt, score all of them, and use the group-relative advantage. The oracle sits INSIDE the update.

GPU-hours per iteration, reconstructed from artifact mtimes

